

Report on *Quantum Variational Algorithms on Adiabatic Quantum Computing Devices*

Major points

1. **Correct the Hamiltonians and enforce one convention throughout.** Equation (2) writes the field term of the TFIM as $-h \sum_i \sigma_i^z$, which commutes with the Ising interaction and therefore does not define a transverse-field model. It should be, for the convention used elsewhere,

$$H_{\text{TFIM}} = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x.$$

This correction affects the physical interpretation, the local-energy estimator, the exact solution, and all statements about the ordered, critical, and disordered regimes. The report should also state whether the Heisenberg Hamiltonians use Pauli matrices σ or spin operators $\mathbf{S} = \sigma/2$, because the energies differ by a factor of four. Boundary conditions and the normalization of J and h must be fixed once and used in every figure.

2. **Reframe the central research question and align it with the evidence.** The introduction promises a systematic assessment of D-Wave as a Gibbs sampler on computationally difficult problems. The results instead contain three partly separate studies: sampler fidelity on small RBMs, representational limits of positive RBMs, and hardware embedding strategies. The report should state one primary question, for example: *How do sampler bias, temperature calibration, and hardware topology affect the accuracy and resource cost of RBM variational Monte Carlo?* Each result can then be tied to one error source. If the principal claim remains about D-Wave Gibbs sampling, the report needs direct distribution-level measurements from the quantum processor over a controlled family of instances, not only training curves and embedding examples.
3. **Define the RBM wave function and the Ising sampling problem rigorously.** Equations (13)–(16) reuse Φ for different objects, omit the hidden-state summation index in places, and leave the sign convention of the RBM energy unclear. A clean definition is

$$E_\theta(v, h) = -a^\top v - b^\top h - v^\top W h, \quad p_\theta(v, h) = Z^{-1} e^{-E_\theta(v, h)},$$

followed by $|\psi_\theta(v)|^2 = p_\theta(v) = \sum_h p_\theta(v, h)$. For a positive-amplitude state,

$$\psi_\theta(v) \propto e^{a^\top v/2} \prod_j \left[2 \cosh \left(b_j + \sum_i W_{ij} v_i \right) \right]^{1/2}.$$

The report should explain exactly what is sent to each sampler, how visible and hidden spins are encoded, whether samples are drawn from the joint distribution and then marginalized, and how hardware ranges rescale a , b , and W .

4. **Resolve the effective temperature notation and validate conditional expectation matching.** Equation (18), $\beta_{\text{eff}} = 1/\beta_x$, is not generally true without a precise definition of β_x . If the programmed energy is $E_{\text{in}} = \alpha E_\theta$ and the device has inverse temperature β_{hw} , then the distribution relative to the unscaled model has $\beta_{\text{eff}} = \alpha\beta_{\text{hw}}$. A reciprocal law follows only if $\beta_x = 1/\alpha$ and β_{hw} is held fixed. This should be derived explicitly. The CEM estimate should then be validated against exact maximum-likelihood or total-variation estimates for several sizes, fields, random parameter sets, and training iterations. Report bias, variance, confidence intervals, and the additional sampling cost of clamping visible variables. One illustrative fit at $N = 16$ is not sufficient to establish reliability during a full optimization.
5. **Use a statistically fair benchmarking protocol.** The new time-to-accuracy plot selects the best configuration for each method and size. This gives an optimistic estimate and can favor methods with a larger tuning budget. Define a tuning set and a disjoint evaluation set, apply the same number of tuning trials to every method, and report the selected configuration before evaluating the test instances. Replace “best of three seeds” by the median or mean over independent seeds, with confidence intervals and all failures retained as censored observations. The benchmark should include multiple Hamiltonian instances at each size, not only one trained trajectory. The stopping rule based on a ten-iteration causal rolling mean must specify whether the threshold must remain satisfied for subsequent iterations and how runs that never reach the threshold are handled.
6. **Separate sampling time, optimization time, and end-to-end time.** The reported cumulative sampling time omits or does not clearly identify parameter updates, JAX compilation, embedding, device programming, network latency, readout, temperature estimation, and hyperparameter search. For an applied comparison, provide at least three clocks: sampler kernel time, device-access time, and end-to-end wall time. The same accounting boundary must be used for all methods. For D-Wave, distinguish anneal time from programming, thermalization, and readout time. For an FPGA or GPU, include data transfer and host-control overhead. A second analysis may report amortized costs when an embedding or compiled kernel is reused.
7. **Treat the energy comparison as preliminary until power is measured.** Several curves in the energy consumed to epsilon are marked “assumed” and use constant power rather than telemetry. Such curves are useful for sensitivity analysis, but they cannot establish measured energy efficiency. The report should state the assumed power values, their source, whether they refer to accelerator, host, or wall power, and how idle power is treated. The definitive experiment should measure synchronized wall-plug or board-level power, subtract a clearly defined baseline if appropriate, integrate power over the same end-to-end interval, and propagate uncertainty. For remote quantum hardware, the scope should be explicit: QPU access energy, facility energy, and client-side energy are different quantities.
8. **Provide complete hardware and software metadata.** The report must identify the exact D-Wave solver names, processor generations, access dates, active-qubit and coupler counts, anneal schedules, anneal times, numbers of reads, programming and readout settings, gauge transformations, chain strengths, chain-break treatment, embeddings, and calibration controls. The FPGA model, clock frequency, numerical precision, implemented update rule, number of parallel trajectories, and power-measurement method are also required. For every classical baseline, give the software version, processor or accelerator, thread count, compiler, random-number generator, temperature schedule, burn-in, thinning, and initialization. Without these details, the reported times and distributions cannot be reproduced.
9. **Disentangle ansatz error, finite-sample error, sampler bias, and hardware distortion.** The current presentation sometimes attributes an energy error directly to the sampler although sparsity changes the variational family itself. A useful decomposition is: (i) exact optimization

of the chosen ansatz or the best value found with exact sampling; (ii) finite-sample error from ideal independent samples; (iii) Markov-chain correlation and non-equilibrium sampler bias; (iv) embedding and control errors; and (v) optimization instability. For small systems, each term can be measured by replacing one component at a time. This ablation is necessary to interpret Fig. 11, where the sparse model loses expressive power before any QPU noise is introduced.

10. **Strengthen the analysis of sparse and parallel embeddings.** A single greedy sparse mask does not establish a general accuracy–connectivity relation. Compare several masks at the same number of edges, including random, degree-balanced, hardware-native, and optimization-aware masks. Control for parameter count and hidden-unit number. For parallel embeddings, measure the distribution and energy obtained by each physical copy, pairwise correlations between copies, spatial calibration bias, and the effective number of independent samples. Simultaneous copies may share low-frequency noise or control errors, so the nominal multiplication of sample count should not be assumed to produce the same multiplication of independent information.
11. **Correct and verify the exact-solution section.** The Jordan–Wigner expression in Eq. (30) has mismatched parentheses, and the dispersion and ground-state-energy formulas in Eqs. (31)–(35) appear to miss convention-dependent factors of two or one half. Derive the formulas from the corrected Hamiltonian and verify them numerically against sparse exact diagonalization for several small N , both parity sectors, and values of h/J . The statement of $O(N)$ complexity should be limited to evaluating the uniform-chain ground-state energy once the allowed momenta are known; it does not apply to constructing the full state or arbitrary observables. The exact-diagonalization description should also clarify that the Hamiltonian is applied as a sparse operator rather than stored as a dense $2^N \times 2^N$ matrix.
12. **Treat the sign structure as a scope boundary rather than a secondary detail.** A real positive RBM cannot represent a generic frustrated ground state. The Marshall transformation is exact for the unfrustrated bipartite antiferromagnet but does not generally remove the sign structure introduced by next-nearest-neighbor frustration. The report should define the “sign impurity” plotted in Fig. 9, distinguish exact sign information from a heuristic correction, and compare with at least one phase-capable ansatz, such as a complex RBM or a separate amplitude and phase network. The very large errors beyond $J_2/J_1 \simeq 0.5$ should be presented as a limitation of the chosen state representation, not as a failure of annealing-based sampling.
13. **Add a discussion, conclusion, and reproducibility section.** The report currently ends with implementation details and an appendix derivation, without a synthesis of the findings. Add a discussion that states what is established, what remains preliminary, and which observations depend on small systems. The conclusion should avoid a quantum-advantage claim and instead summarize the practical conditions under which annealing samplers are useful. Release the code, configuration files, raw samples, embeddings, power traces, and scripts that generate every figure. Include a table mapping each figure to its data file, random seeds, and command line.

Minor points and presentation corrections

1. The title contains “Adabatic”; use “Adiabatic.” More precisely, because the study concerns D-Wave processors, “quantum annealing devices” is preferable to the broader term “adiabatic quantum computing devices.”
2. Add an abstract, a table of contents if required by the degree program, and a short list of the report’s original contributions. Page 2 is blank and should be removed unless it is intentional for printing.
3. On page 3, replace “Since the development of the quantum computers” by “Since the development

of programmable quantum processors.” Correct “researcher” to “researchers,” “fast” to “faster,” “possible” to “possibly,” and “conventional” to “conventional.”

4. The statement about a 2025 beyond-classical D-Wave result needs a precise citation and a balanced description of the problem, observable, and classical-comparison boundary. It should not be used as a general claim that the device is faster for optimization or sampling.
5. In Eq. (1), state the assumptions behind time to solution: independent runs with constant single-run success probability and $0 < P_{\text{success}} < 1$. Explain how zero observed successes are treated.
6. Replace “limiting connectivity” by “limited connectivity” and “lined out” by “arranged.” Define logical and physical qubits before Fig. 2.
7. Equation (8) contains a stray equality sign after the sum. Use $\sum_{i < j}$ rather than summing over all ordered pairs unless a factor of one half is included.
8. The adiabatic-theorem discussion is too absolute for an open-system annealer. State that the hardware implements a finite-time, open-system annealing process and that final samples need not be ground states of the programmed Hamiltonian.
9. In Fig. 3, correct “orange” to “orange.” The schedule curves should be identified as schematic or tied to a named processor calibration.
10. “Boltzman” should be “Boltzmann” throughout. Other recurring corrections include “temeperature” → “temperature,” “sqaured” → “squared,” “visible” → “visible,” “auxillary” → “auxiliary,” “ressources” → “resources,” “caluclation” → “calculation,” “differentation” → “differentiation,” “annihilation” → “annihilation,” and “Bogugliev” → “Bogoliubov.”
11. The report calls the method VQE, but the implemented algorithm is more accurately described as neural-network variational Monte Carlo. Reserve “VQE” for the general variational principle or explain the terminology.
12. In Sec. 4.1, the reference to “Figure 3a” should be “Fig. 5(a).” Use “total variation distance,” not “total variational distance.”
13. The dashed “sampling floor” is not a strict lower bound. It is a finite-sample reference, normally estimated from repeated independent samples drawn from the exact distribution. State how many repetitions were used and show an uncertainty band.
14. Sweeps for single-spin methods and steps for a parallel oscillator method are not equivalent computational effort. Keep the algorithmic-effort plot, but add wall-time and effective-sample-size comparisons.
15. The claim that Metropolis and Gibbs are “theoretically near-optimal” should be replaced by the empirical statement that they approach the finite-sample reference for the tested $N = 8$ distributions.
16. Define burn-in, chain initialization, thinning, autocorrelation time, and effective sample size for every Markov-chain result. Ten thousand correlated samples are not equivalent to ten thousand independent samples.
17. In the SR equations, use complex conjugation for the general case, $S_{kk'} = \langle O_k^* O_{k'} \rangle - \langle O_k^* \rangle \langle O_{k'} \rangle$, and state explicitly when all parameters and observables are real. Give the regularization schedule, conjugate-gradient tolerance, and stopping criterion.

18. Figure 8 should state the boundary conditions, J , sample autocorrelation treatment, and the exact method used for the dashed correlations. The reported energy error should include uncertainty over independent training runs.
19. Figure 9 uses percentages above 100%. Define the denominator and explain why this relative error is informative when the exact energy is negative. An absolute error per site is often easier to interpret.
20. Figure 11’s caption contains incomplete text (“kept equal t” and “lowestllows”). The vertical “native hardware floor” needs a definition in the main text.
21. Figure 12 compares $n_{\text{parallel}} = 1, 3, 5$ in the curve but shows four copies in the layout. Explain this difference. Replace “best of 3 seeds” with a distribution over seeds.
22. The JAX section is useful but too tutorial-like relative to the scientific results. Condense general explanations of pure functions, JIT, and `vmap`; retain only implementation choices that affect scaling or reproducibility.
23. Replace the blog citation for Metropolis–Hastings and the lecture-note citation for the TFIM by primary papers or standard textbooks. Every web reference should include author, title, version, year, and a stable identifier.
24. Use one spelling convention consistently (“optimization” or “optimisation”) and one notation for equation and figure references. Avoid “we will not go into details”; either provide the needed detail or cite a source.
25. Improve figure readability by enlarging labels, using color-blind-safe markers, and placing all panel labels consistently. Captions should define every symbol and state sample counts, seeds, and error-bar meaning.